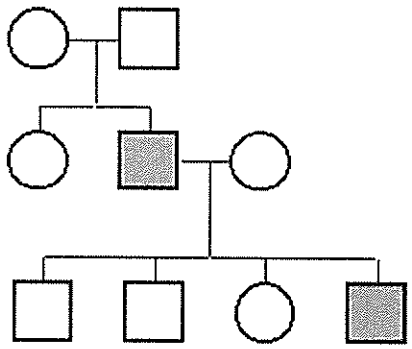
Pedigree Analysis

Introduction

A pedigree is a diagram of family relationships that uses symbols to represent people and lines to represent genetic relationships. These diagrams make it easier to visualize relationships within families, particularly large extended families. Pedigrees are often used to determine the mode of inheritance (dominant, recessive, etc.) of genetic diseases. A sample pedigree is below.



In a pedigree, squares represent males and circles represent females. Horizontal lines connecting a male and female represent mating. Vertical lines extending downward from a couple represent their children. Subsequent generations are therefore written underneath the parental generations and the oldest individuals are found at the top of the pedigree.

If the purpose of a pedigree is to analyze the pattern of inheritance of a particular trait, it is customary to shade in the symbol of all individuals that possess this trait.

In the pedigree above, the grandparents had two children, a son and a daughter. The son had the trait in question. One of his four children also had the trait.

In the exercises below, assume that the trait in question is a genetic disease or abnormality. We will learn patterns of inheritance that have the following modes of inheritance:

autosomal dominant
autosomal recessive
X-linked recessive

Developing Conclusions About Different Modes of Inheritance

Autosomal Dominant

1. The pedigree below is for a genetic disease or abnormality. We do not yet know if it is dominant or recessive. We will determine if it is possible that the trait is autosomal dominant. If the trait were dominant, we could use the following designations:

A = the trait (a genetic disease or abnormality, dominant)
a = normal (recessive)

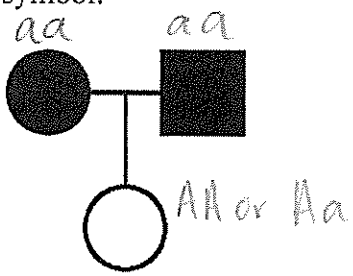
c) In conclusion, can two individuals that have an autosomal dominant trait have unaffected children? Write your conclusion on the answer sheet (question 2). Yes, if they are both Heterozygous Carriers

Autosomal Recessive

3. We will determine if the pedigree below can be for a trait that is autosomal recessive. Use the following designations:

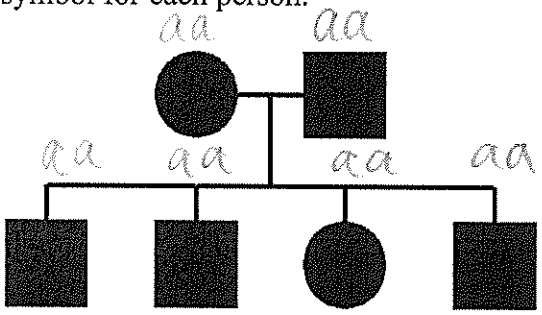
A = normal
a = the trait (a genetic disease or abnormality)

a) Assuming that the trait is recessive, write the genotype of each individual next to the symbol.



b) Is it possible that the pedigree above is for an autosomal recessive trait? No

c) Assuming that the pedigree below is for a recessive trait, write the genotype next to the symbol for each person.



d) Is it possible that this pedigree is for an autosomal recessive trait? yes

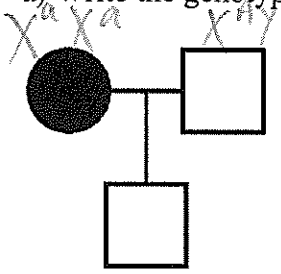
e) If a trait is autosomal recessive, what can you conclude about the children if both parents are affected? Write your conclusion on the answer sheet (question 3). Homozygous recessive all have it

4. We will determine if the pedigree below can be for a trait that is autosomal recessive. Use "A" and "a" as you did for the previous example.

a) Write the genotype of each individual next to the symbol.

X^A = normal
X^a = the trait (a genetic disease or abnormality)
Y = Y chromosome (males only)

a) Write the genotype of each individual next to the symbol.

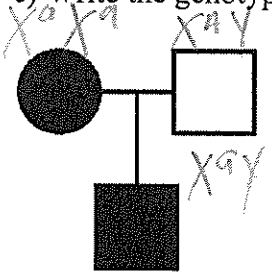


	X^a	X^a
X^A	X^AX^a	X^AX^a
Y	X^aY	X^aY

100% All males would have trait

b) Is it possible that the pedigree above is for an X-linked recessive trait? NO

c) Write the genotype next to the symbol for each person in the pedigree below.

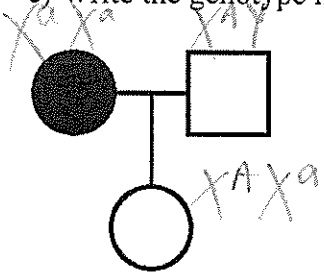


	X^a	X^a
X^A	X^AX^a	X^AX^a
Y	X^aY	X^aY

all females have it
all males have it

d) Is it possible that this pedigree is for an X-linked recessive trait? ✓ YES

e) Write the genotype next to the symbol for each person in the pedigree below.

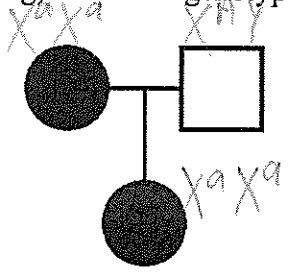


	X^A	Y
X^a	X^AX^a	X^aY
X^a	X^AX^a	X^aY

All males would have it
All females Carriers

f) Is it possible that this pedigree is for an X-linked recessive trait? Yes

g) Write the genotype next to the symbol for each person in the pedigree below.



	X^a	X^a
X^A	X^AX^a	X^AX^a
Y	X^aY	X^aY

All females would be Carriers but not show trait

h) Is it possible that this pedigree is for an X-linked recessive trait? NO

e) Do the conclusions that you made for autosomal recessive traits apply to X-linked recessive traits? Write your conclusion on the answer sheet (question 9).

Yes.

10a. If a genetic disease is X-linked recessive, what is the phenotype of a female that has only one disease allele (X^a)? $X^A X^a$ (Heterozygous X linked)

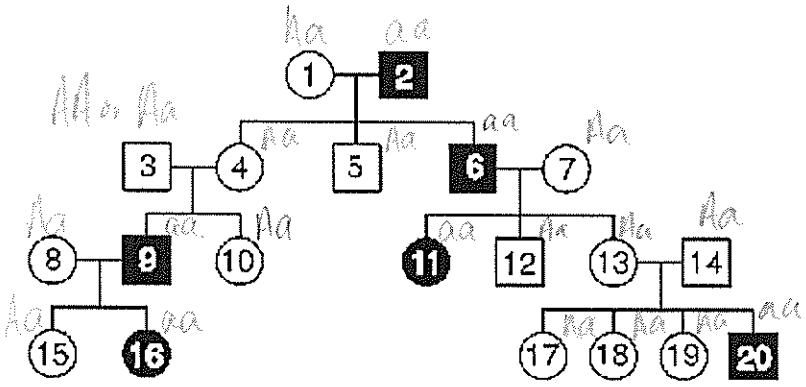
b. What is the phenotype of a male with one disease allele? Has the genetic disorder

c. What can you conclude about the number of males that would have the disease compared to the number of females? Write your conclusion on the answer sheet (question 10).

Application of Pedigree Analysis

The conclusions about inheritance (above) can be used to help analyze pedigrees. For each pedigree below, tell if the trait can be autosomal dominant, autosomal recessive, and X-linked recessive. If the pedigree cannot fit a mode of inheritance, tell why.

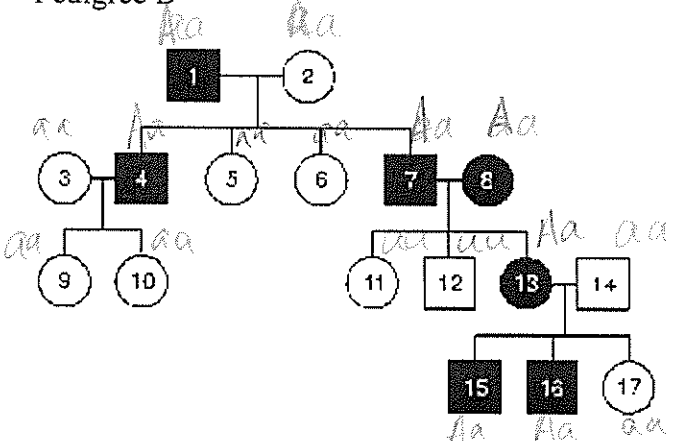
Pedigree A



Autosomal recessive or sex linked recessive

* Where is the aa?

Pedigree B



Autosomal dominant

Questions 10 through 15- Answer with either "yes" or "no." If you answer no, explain why.

11) Can pedigree A be autosomal dominant? To answer this question, go back and look at your conclusion for autosomal dominant traits (question 1e). NO